Entity Resolved Knowledge Graphs: The Foundation for Effective GraphRAG

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CLAIR SULLIVAN
& ASSOCIATES

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github.com/cj2001/
odsc_east_2026

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Outline

- Terminology: graphs and knowledge and entities, oh my!
- Why is entity resolution (ER) hard
- Why does ER matter for RAG?
- Coding!
- Story time: the real-world solution to this ERKG data
- Why not just use an LLM for ER?
- Conclusion

Some definitions

- Graph
- Knowledge graph (KG)
- Entity resolution (ER)
- Entity-resolved knowledge graph (ERKG)

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What is an entity?

- Distinct thing
  - Person
  - Place
  - Organization
  - Object
- Think of them like rows in a table
- Defined by a combination of distinguishing characteristics
  - Some characteristics are more distinguishing than others!

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Consider the following scenarios...

Five different people, each with a single transaction

A single person with five different transactions

Why does it matter?

- Counting records is the most basic task of a data scientist
- If you can't count, you can't predict
- If you can't count, you can't quantify
- Enables things like:
  - Good news / bad news
  - Red light / green light
  - To market to them or not
  - Benign / threat

Hot take: regex is not the answer

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Names are complicated

Johnathan Edward Smith	Jonathan Edward Smith	Jonathon Edward Smith	Jonothan Edward Smith	Jonatan Edward Smith	John Edward Smith	Jon Edward Smith	Johnny Edward Smith
Jonny Edward Smith	Jack Edward Smith	Jono Edward Smith	Nate Edward Smith	Johnathan Ed Smith	Jonathan Ed Smith	Jonathon Ed Smith	Johnathan Eddie Smith
Jonathan Eddie Smith	Jon Eddie Smith	Johnathan Ted Smith	Johnny Ted Smith	Johnathan Teddy Smith	Johnathan Ned Smith	Johnathan E. Smith	Jonathan E. Smith
Jonathon E. Smith	Jon E. Smith	John E. Smith	Jack E. Smith	Jono E. Smith	Johnny E. Smith	J. Edward Smith	J. E. Smith
J.E. Smith	J. Smith	Jon Smith	John Smith	Johnny Smith	Jack Smith	Jono Smith	Eddie Smith
Ted Smith	Teddy Smith	Ned Smith	Nate Smith	Johnathan Edward Smithe	Jonathan Edward Smithe	Jonathon Edward Smithe	Jon E. Smithe
Johnny E. Smithe	Johnathan E. Smyth	Jonathan E. Smyth	Jon Smyth	Johnny Smyth	John Smyth	Jack Smyth	Jono Smyth
J. Edward Smyth	J. E. Smyth	J.E. Smyth	Johnathan Edward Smythe	Jonathan Edward Smythe	Jonathon E. Smythe	Jon Smythe	John Smythe
Johnny Smythe	J. E. Smythe	Ted Smythe	Eddie Smythe	Nate Smythe			

Addresses are REALLY complicated

123 Main, Apt 1A, Las Vegas, NV, 89111	123 Main, #1A, Las Vegas, Nevada, 89111	123 Main St, Apt 1A, Las Vegas, NV, 89111	123 Main St, #1A, Las Vegas, Nevada, 89111	123 Main St., Apt 1A, Las Vegas, NV, 89111	123 Main St., #1A, Las Vegas, Nevada, 89111	123 Main Street, Apt 1A, Las Vegas, NV, 89111	123 Main Street, #1A, Las Vegas, Nevada, 89111
123 Main, Apt 1A, Las Vegas, Nevada, 89111	123 Main, Apartment 1A, Las Vegas, NV, 89111	123 Main St, Apt 1A, Las Vegas, Nevada, 89111	123 Main St, Apartment 1A, Las Vegas, NV, 89111	123 Main St., Apt 1A, Las Vegas, Nevada, 89111	123 Main St., Apartment 1A, Las Vegas, NV, 89111	123 Main Street, Apt 1A, Las Vegas, Nevada, 89111	123 Main Street, Apartment 1A, Las Vegas, NV, 89111
123 Main, Apt. 1A, Las Vegas, NV, 89111	123 Main, Apartment 1A, Las Vegas, Nevada, 89111	123 Main St, Apt. 1A, Las Vegas, NV, 89111	123 Main St, Apartment 1A, Las Vegas, Nevada, 89111	123 Main St., Apt. 1A, Las Vegas, NV, 89111	123 Main St., Apartment 1A, Las Vegas, Nevada, 89111	123 Main Street, Apt. 1A, Las Vegas, NV, 89111	123 Main Street, Apartment 1A, Las Vegas, Nevada, 89111
123 Main, Apt. 1A, Las Vegas, Nevada, 89111	123 Main, Unit 1A, Las Vegas, NV, 89111	123 Main St, Apt. 1A, Las Vegas, Nevada, 89111	123 Main St, Unit 1A, Las Vegas, NV, 89111	123 Main St., Apt. 1A, Las Vegas, Nevada, 89111	123 Main St., Unit 1A, Las Vegas, NV, 89111	123 Main Street, Apt. 1A, Las Vegas, Nevada, 89111	123 Main Street, Unit 1A, Las Vegas, NV, 89111
123 Main, #1A, Las Vegas, NV, 89111	123 Main, Unit 1A, Las Vegas, Nevada, 89111	123 Main St, #1A, Las Vegas, NV, 89111	123 Main St, Unit 1A, Las Vegas, Nevada, 89111	123 Main St., #1A, Las Vegas, NV, 89111	123 Main St., Unit 1A, Las Vegas, Nevada, 89111	123 Main Street, #1A, Las Vegas, NV, 89111	123 Main Street, Unit 1A, Las Vegas, Nevada, 89111
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123 Main, Apartment 1A, Las Vegas, Nevada, 89111	123 Main St, Apt. 1A, Las Vegas, NV, 89111	123 Main St, Apartment 1A, Las Vegas, Nevada, 89111	123 Main St., Apt. 1A, Las Vegas, NV, 89111	123 Main St., Apartment 1A, Las Vegas, Nevada, 89111	123 Main Street, Apt. 1A, Las Vegas, NV, 89111		

Different character sets

جوناثان إدوارد سميث

A شارع مين، شقة 1 123

لاس فيغاس، نيفادا 89111

Cultural variations

Argentina: Piedras 623, Piso 2, depto 4, C1070AAM, Capital Federal

Belgium: Rue du Vivier 7C bte 5, 1000 Bruxelles, BELGIQUE

India: e-506 street number 78, uttam viharblock d, uttam nagar, bindapur, 110059, new delhi

New Zealand: 43 Vogel Street, Roslyn, Palmerston North 4414

Ukraine: vul. Shevchenka, bud. 17, m. Bila Tserkva, Kyivs'ka obl., 09117, UKRAINE

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Your RAG is only as good as your ER!

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Date of Birth: 9-22-61
Address: 123 Main St, Las Vegas, NV 89111
Phone: 555-654-1234
Email: j.smith@domain.com

Johnathan Smith

Date of Birth: 16.3.85
Address: 456 1st St, Las Vegas, NV 89111
Phone: 555-123-4567
Email: john.smith123@outlook.com

Jack Smith

Date of Birth: 16-Mar-1985
Address: 123 Main St #2B, Las Vegas, NV 89111
Phone: 555-654-1234
Email: j.smith@domain.com

John Smith

Date of Birth: Sept. 22, 1961
Address:
Phone:
Email: smith.john@gmail.com
ID#: 987654321

Dr. J. Smith

ODSC AI

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Phone: 555-654-1234
Email: j.smith@domain.com

Johnathan Smith

Date of Birth: 16.3.85
Address: 456 1st St, Las Vegas, NV 89111
Phone: 555-123-4567
Email: john.smith123@outlook.com

Jack Smith

"Tell me about Jonathan Smith"

Date of Birth: 16-Mar-1985
Address: 123 Main St #2B, Las Vegas, NV 89111
Phone: 555-654-1234
Email: j.smith@domain.com

John Smith

Address:
Phone:
Email: smith.john@gmail.com
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Dr. J. Smith

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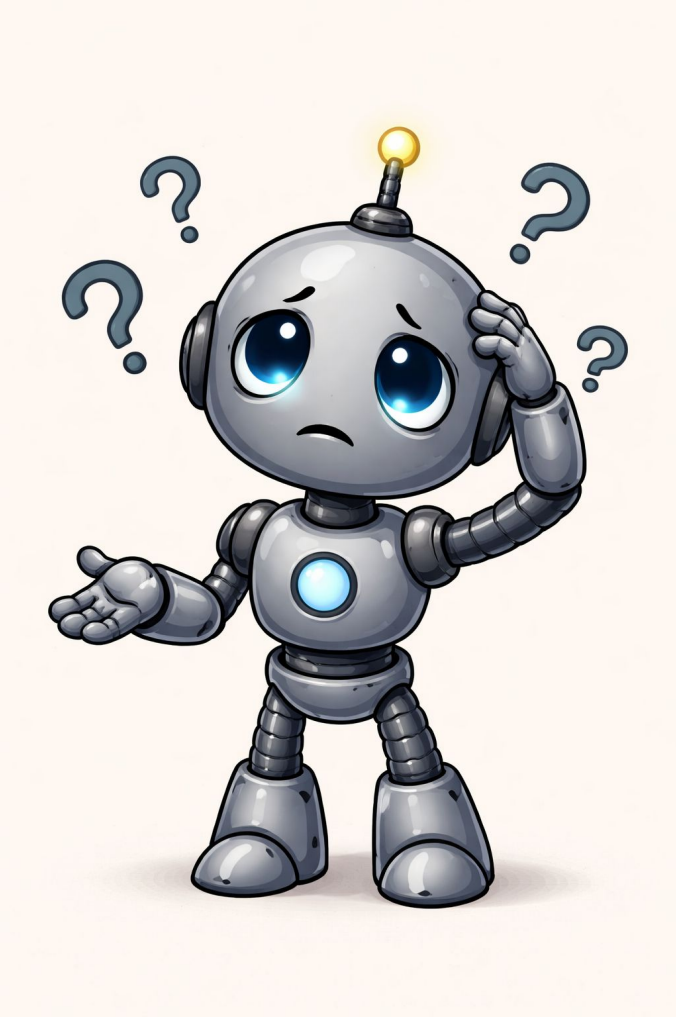

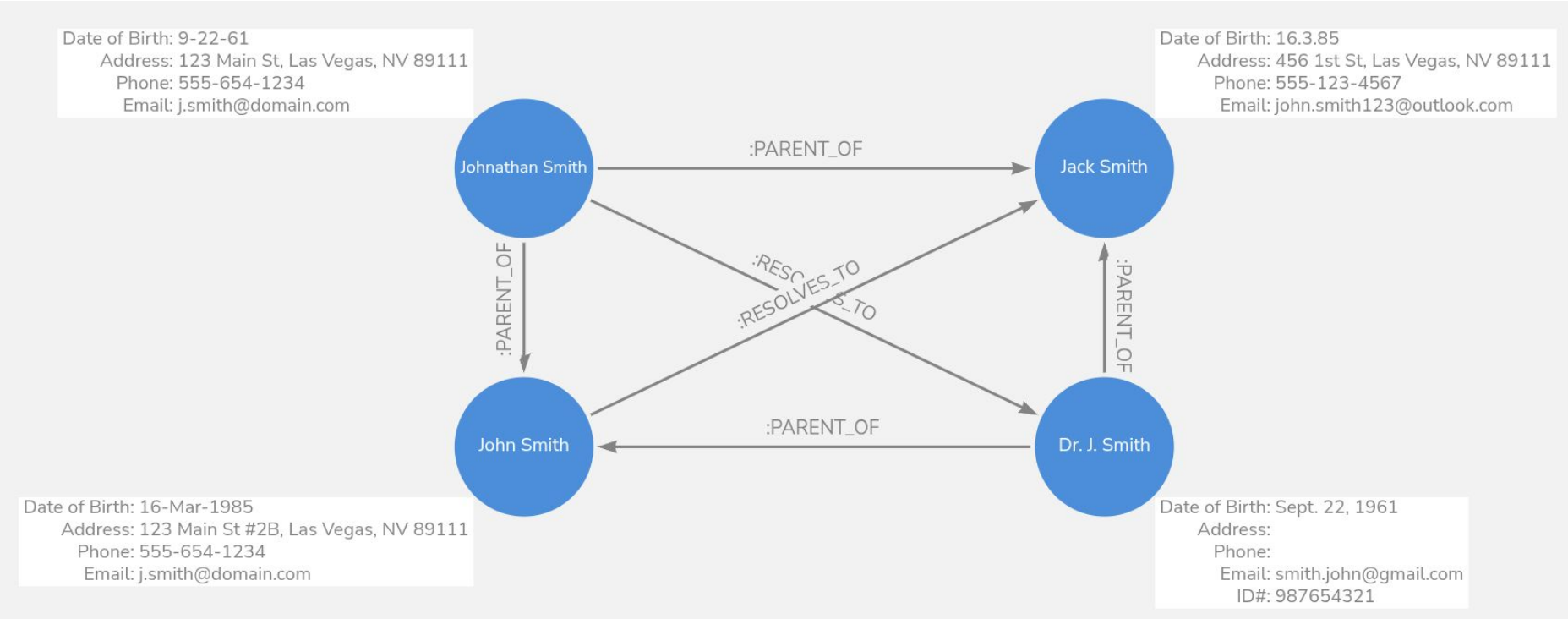
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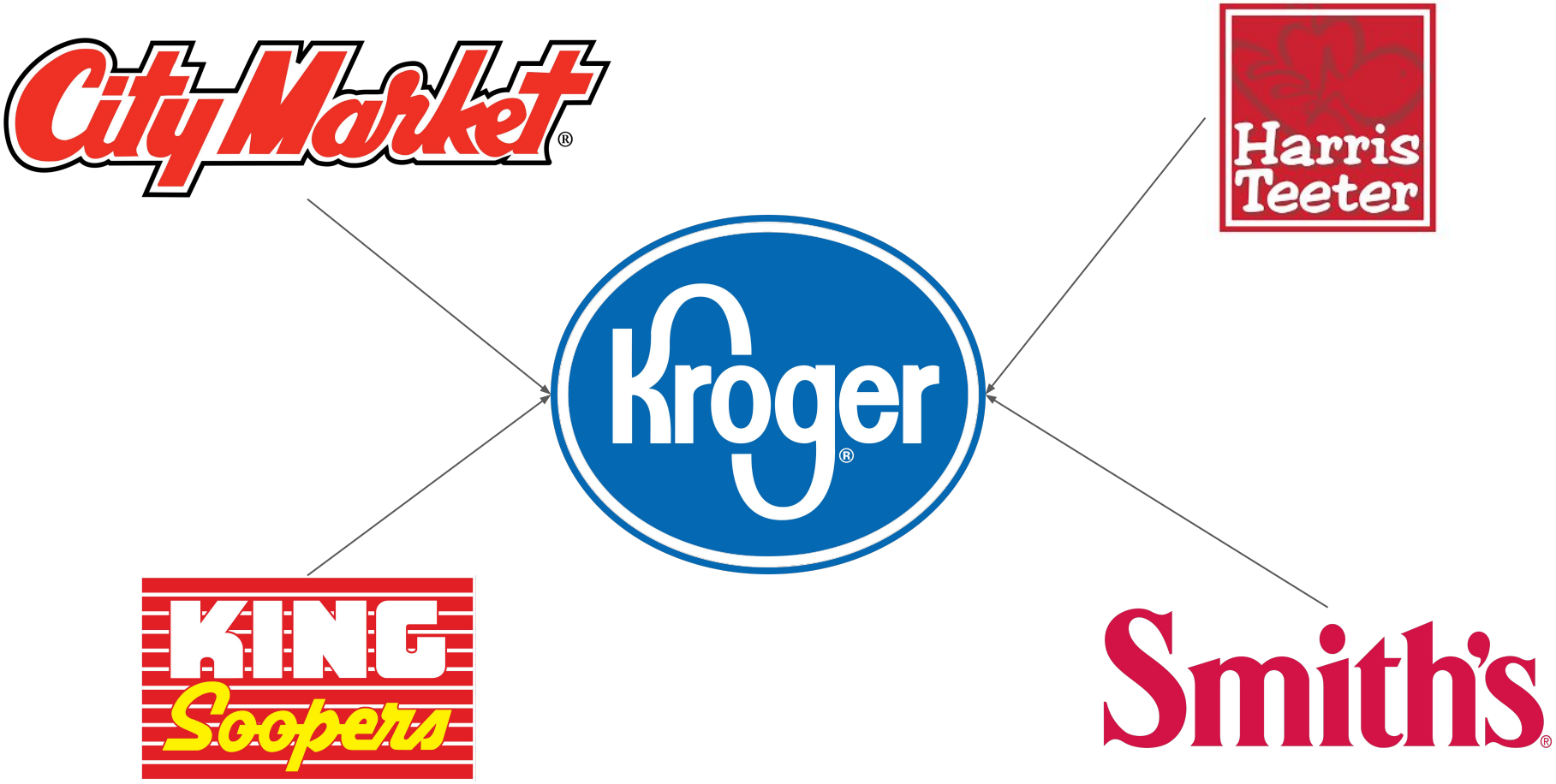
Email: john.smith123@outlook.com
Johnathan Smith
:PARENT_OF
Jack Smith
:PARENT_OF
:RESOLVES_TO
:RESOLVES_TO
:PARENT_OF
John Smith
:PARENT_OF
Dr. J. Smith
Date of Birth: 16-Mar-1985
Address: 123 Main St #2B, Las Vegas, NV 89111
Phone: 555-654-1234
Email: j.smith@domain.com
Date of Birth: Sept. 22, 1961
Address:
Phone:
Email: smith.john@gmail.com
ID#: 987654321


City Market
Harris Teeter
Kroger
King Soopers
Smith's

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odsc_east_2026

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The code

- 00: Verifies workshop environment is properly configured and running
- 01: Conduct ER on a known dataset
- 02: Perform ER on the unknown dataset
- 03: Explore the output of Senzing on the unknown dataset
- 04: Create and visualize the ERKG
- 05: Traditional RAG approach (no ER, no graphs)
- 06: Traditional GraphRAG approach (no ER)
- 07: Vectorize ER data
- 08: Full ERKG RAG

The truth dataset

- Curated demo data with known ground truth
- 159 records across 3 data sources
  - CUSTOMERS (120 records): contains primary subjects of interest (customers, employees, vendors) and intentionally includes duplicates, name variations, and address changes to test resolution accuracy
  - REFERENCE (22 records): provides external enrichment data such as demographics, past addresses, contact methods for people, and firmographics/corporate structure/ownership for companies
  - WATCHLIST (17 records): represents entities to screen against, such as known fraud actors or sanctioned parties

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OpenSanctions and Open Ownership

- **OpenSanctions** aggregates sanctions lists, watchlists, and politically exposed persons (PEPs) from over 100 sources worldwide
- **Open Ownership** provides structured data on corporate beneficial ownership, meaning who actually owns or controls a company, sourced from national registries across multiple countries
- Both provide data in Senzing JSONL format
- Data subset for today
  - Open Ownership: 316 records
  - OpenSanctions: 24 records
  - **A real story to be told!**

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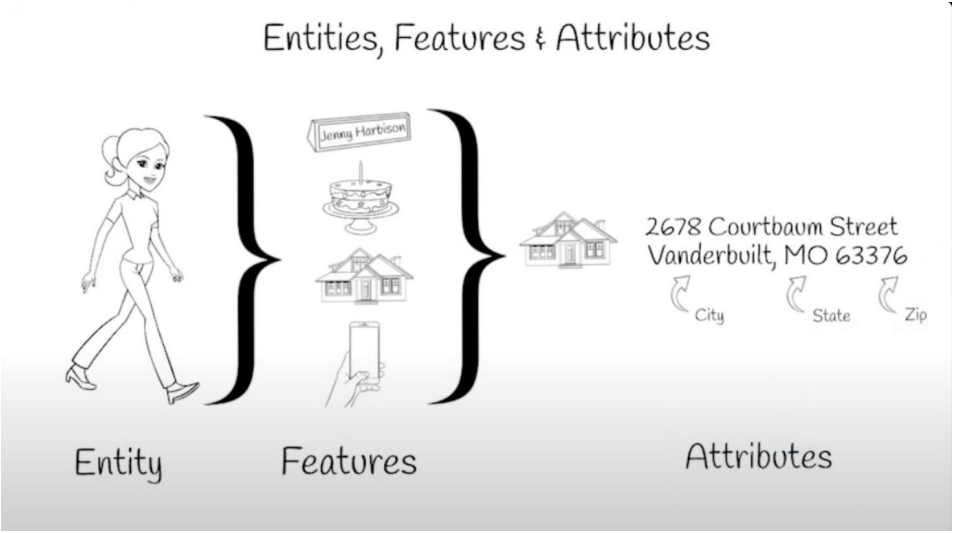
Senzing Entity Specification

- Defines rules for how to map source fields to Senzing target fields
- Standardizes diverse data sources into a common format for ER



https://senzing.com/docs/entity_specification/

The solution!

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From pizza delivery driver to multi-unit franchisee

Posted: 12 Feb 2019 | 2 minutes read

Share:

Former Papa John's delivery driver Abassin Badshah has opened his second franchise in partnership with the American brand in Bromley, London.

Abassin, who joined Papa John's in 2002, now employs around 30 people. He bought the Catford Papa John's franchise in 2013.

He explains: "I worked for Papa John's when I was a student in business.

"By day I would attend lectures, then in the evenings and at weekends I would deliver pizza. I was successful at my studies, but also within Papa John's and gradually worked my way up to supervisor. The

https://www.what-franchise.com/news/from-pizza-delivery-driver-to-multi-unit-franchisee

https://www.pointfranchise.co.uk/news-papa-john-s-delivery-driver-climbs-the-ladder-to-franchisee

Papa John's delivery driver climbs the ladder to franchisee

Sophie Cole, writer

Published at 18/02/2019 , Updated on 07/10/2020, Reading time: 1 min

Papa John's has shared the incredible story of how one of its former delivery drivers has become a successful franchisee.

Many of us dream of progressing up the career ladder and claiming the top spot one day, but a franchisee of Papa John's has made that dream a reality.

Abassin Badshah began his career in 2002 as a delivery driver for his local Papa John's branch. It was a part-time job for him while he studied business at university, but the ambitious worker gradually progressed to the role of supervisor at the Catford branch of the pizza franchise. Badshah explains what motivated him to look into buying his very own franchise.

"I knew the pizza was amazing. Papa John's concept of: 'BETTER INGREDIENTS. BETTER PIZZA.' means we only use the finest ingredients and fresh dough which gives it the best taste! It also meant there was a huge demand and customers were really hungry for what we offered! I realised becoming a franchisee and having my own Papa John's would be a great business opportunity."

He took the plunge six years ago and purchased the Catford branch. Explaining the challenges he faced at that time and when he decided to open his second franchise, Badshah says:

Former Pizza Delivery Driver Opens Second Papa John's

11 February 2019

Leading pizza franchise, Papa John's, has announced franchisee Abassin Badshah has opened in Bromley, London. Former Papa John's delivery driver Abassin who joined Papa John's in 2002, now employs more than 30 local people across his two outlets in Catford and Bromley.

"I worked for Papa John's when I was a student in business," explains Abassin. "By day I would attend lectures then in the evenings and at weekends, I would deliver pizza! I was successful at my studies but also within Papa John's and gradually worked my way up to supervisor. The experience gave me the grounding to understand how the business really worked.

"I knew the pizza was amazing," continues Abassin. "Papa John's concept of: "BETTER INGREDIENTS. BETTER PIZZA." means we only use the finest ingredients and fresh dough which gives it the best taste! It also meant there was a huge demand and customers were really hungry for what we offered! I realised becoming a franchisee and having my own Papa John's would be a great business opportunity.

https://inside.papajohns.co.uk/former-pizza-delivery-driver-opens-second-papa-johns/

LEGAL | STAFF REPORTER , UK

PUBLISHED: *5 YEARS AGO* | 224 VIEWS

Papa John's franchisee jailed for tax fraud: report

The chain was not subject to the criminal investigation or court proceedings.

A Papa John's franchisee who 'cooked the books' at stores in Catford and Bromley has been jailed for tax fraud.

News Shopper reported that HMRC's Fraud Investigation service said Abassin Badshah, of Quernmore Close, Bromley, under-declared his pizzeria's income, evading a total of £669,000 in Income Tax, National Insurance and Corporation Tax.

The 35-year-old was jailed for four years following an investigation by HM Revenue and Customs (HMRC).

The report added that Badshah suppressed the true sales figures for the Papa John's takeaway he ran at 223 Bromley Road, Bellingham, London, to evade Income Tax and National Insurance.

Papa John's were not subject to the criminal investigation or court proceedings, the report noted.

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Read the full report here.

https://qsrmedia.co.uk/legal/more-news/papa-johns-franchisee-jailed-tax-fraud-report

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GOV·UK

Find and update company information

Companies House does not check the accuracy of the information filed

Sign in / Register

Search for a company or officer

Advanced company search

Abassin BADSHAH

Filter appointments

☐ Current appointments

Total number of appointments 3

Date of birth
May 1985

WELLHANCIA HEALTH CARE LTD (11896001)

Company status	Correspondence address
Dissolved	**31 Quernmore Close, Bromley, United Kingdom, BR1 4EL**

Role	Appointed on
Director	**21 March 2019**

Nationality	Country of residence
British	**United Kingdom**

BARLLOWS SERVICES LTD (11472692)

Company status	Correspondence address
Active	**3 Market Parade, 41 East Street, Bromley, England, BR1 1QN**

Role RESIGNED	Appointed on	Resigned on
Director	**19 July 2018**	**6 April 2021**

Nationality	Country of residence
British	**United Kingdom**

LMAR (GB) LTD (08088204)

Company status	Correspondence address
Active	**223 Bromley Road, London, England, SE6 2PG**

https://find-and-update.company-information.service.gov.uk/officers/uIxILC6PYT4Eo2xAanysKAq4rZc/appointments

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Said Kerimov

2 languages

Article Talk Read Edit View history Tools

From Wikipedia, the free encyclopedia

Said Kerimov (born July 6, 1995) is a Russian businessman who is the son of Russian billionaire and politician Suleyman Kerimov. In 2015, he became a majority shareholder in Polyus Gold, Russia's largest gold producer, when his father transferred shares over to Said.

Said Suleymanovich Kerimov	
Born	July 6, 1995 (age 30) Moscow, Russia
Occupation	Businessman
Father	Suleyman Kerimov

During the 2022 Russian Invasion of Ukraine, Kerimov resigned from the board of Polyus Gold on first week of April 2022, while at the same time his family sold its stake in the company worth $6.3 billion which resulted in the family losing majority shareholder position at Polyus Gold.[1]

Career [edit]

In 2014, Kerimov acquired Russia's cinema chain, Cinema Park from Vladimir Potanin for an estimated $385 million.[2] Cinema Park was sold in 2017 to Alexander Mamut.[3][4]

In 2017, Kerimov took over the control of the bankrupt Makhachkala Airport.[5]

Sanctions [edit]

In January 2023 Japan imposed sanctions on Said Kerimov.[6]

PJSC Polyus [edit]

In 2015, Kerimov became a majority shareholder in Polyus Gold, Russia's largest gold producer, when his father Suleyman transferred shares over to Said.[7][8] In the same year Polyus Gold announced that it was delisting from the London Stock Exchange (LSE) and rebranded as PJSC Polyus.[9][10]

In July 2017, Polyus was re-admitted to the London Stock Exchange.[11] As of end 2020, Kerimov owned about 77% shares of the company.[12] In 2021 he sold part of his stake and gave the remaining to the Fund for support of Islamic foundations.[13]

https://en.wikipedia.org/wiki/Said_Kerimov

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Settlement Agreement between the U.S. Department of the Treasury's Office of Foreign Assets Control and IPI Partners, LLC

12/02/2025

The U.S. Department of the Treasury's Office of Foreign Assets Control (OFAC) today announced a $11,485,352 settlement with IPI Partners, LLC ("IPI") to settle its civil liability for 51 apparent violations of OFAC's sanctions against Russia. In 2017 and 2018, IPI solicited and received investments from Russian oligarch Suleiman Kerimov through a series of legal structures, and continued to maintain those investments for four years after OFAC designated Kerimov on April 6, 2018. The settlement amount reflects OFAC's determination that IPI's conduct was non-egregious and not voluntarily self-disclosed.

For more information, please visit the following Enforcement Release.

https://ofac.treasury.gov/recent-actions/20251202

What if we just did ER with GenAI?

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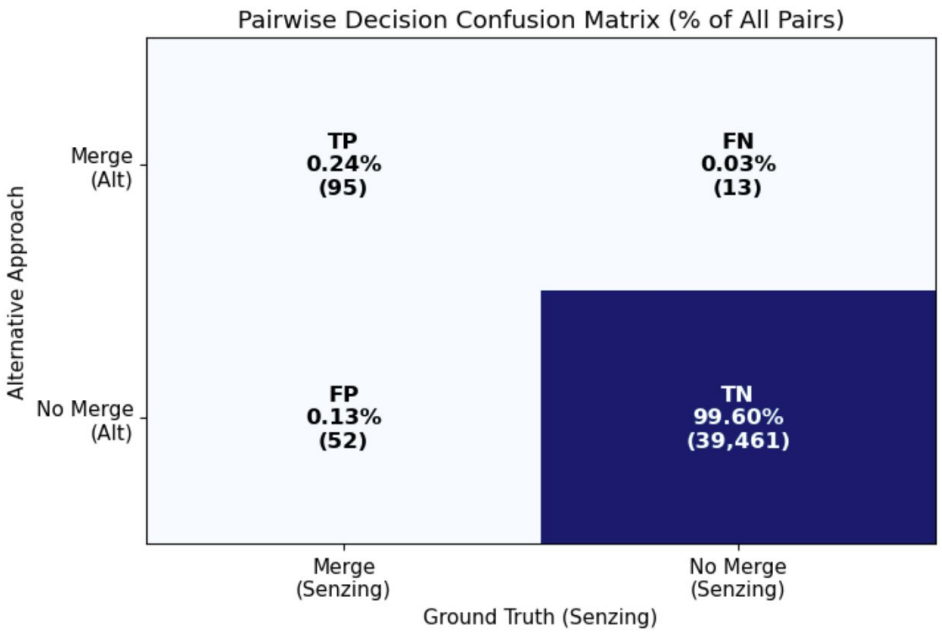
My prompt to Claude

"I am going to provide you with two JSONL files, one for open sanctions and one for open ownership. Your job is to do the entity resolution on these files. I need you to provide me a file that shows which records resolved into which other records. In this file, you must state WHY two records were merged into one. You also need to provide a list of related nodes (i.e. entities that are connected by not merged). I do NOT want you to write the code to do this. I want you to do the entity resolution yourself.

Here are the two files: … "

Pairwise metrics

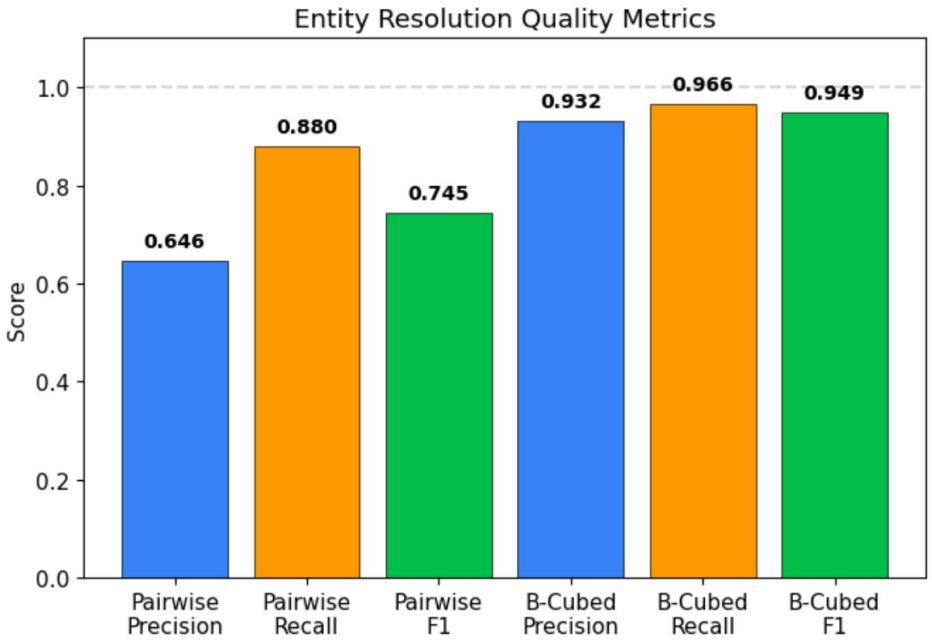
- 282 records = 39,621 possible pairs
  - Senzing said “same entity” for 108 pairs
  - LLM said “same entity” for 147 pairs
- Precision: when the LLM merged two records it was right only 65% of the time
- Recall: of Senzing’s 108 true merges, the LLM found 95 of them
- The F1 score reflects the high recall was dragged down by low precision
- Accuracy is misleading here
  - 99.6% of all pairs a true negatives so you could get near 100% by not merging any records



B-cubed clustering metric

- Evaluates each record individually
  - “Of the records in my cluster, how many actually belong with me?”
- B-cubed precision and recall produce a dramatically higher F1 score than pairwise
- 135 of 282 records are singletons that score a perfect 1.0 on precision and recall, pulling average up

- Bottom line
  - Evaluation metric matters
  - B-cubed: average record is fine
  - Pairwise: a third of merge decisions are wrong
  - Both are true but answer different questions



Other observations and summary

- LLM got roughly 1 in 3 merge decisions wrong
- The biggest merge failure was on overmerging based on unreliable signals
  - Exact address
  - Partial names
- LLM lacked phonetic name matching capabilities
- Relationship detection was almost non-existent
- Senzing processed 282 records in under 10 seconds
  - The LLM took 6 minutes

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Conclusions

- Your RAG is only as trustworthy as the data underneath it
- Doing ER properly is much more difficult than can be solved with regex, edit distance algorithms, clustering, or even by an LLM
- If you cannot accurately identify who or what your entities are, you cannot count them, relate them, or reason about them
- A knowledge graph built on unresolved entities is actively misleading
- ERKGs give your LLM something vector search, RAG, and traditional GraphRAG can not: **REAL context**

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pollEv.com/clairsullivan399

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Thank you!

clair@clairsullivan.com

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& ASSOCIATES